What survives when a paper leaves the lab: re-testing 22 robot-learning papers on a €1,350 rig

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Abstract. We re-tested 37 claims from 22 robot-learning papers on a cheap rig, and 12 of them held. The rig is an SO-101 arm with a Jetson Orin NX, and its parts cost €1,350. Most of the papers were shown in simulation or on far more expensive hardware, so we wanted to know what survives when a claim leaves the lab. For each claim we committed the bar before the test and ran the cheapest baseline first, on the same rows. That baseline matters, because it is often what a method loses to. The failures fall into three classes that we propose: methods too slow for the edge GPU, zero-shot judges that lose to the episode length, and latent planners that lose to the commanded joints. Every number in the paper can be rebuilt from the public repository.

Code and rows: `github.com/ravediamond/paper-register-reproducibility` **Reproduce:** `make check`. Every number carries its source file and protocol commit.

1 Introduction

Most robot-learning papers are shown in simulation or on expensive rigs. Of the 22 papers in this study, 16 were shown in simulation or on a rig above $20k, and 2 on a setup under $5k like ours. A small team that wants to use these methods has to guess which claims still hold on its own hardware. We built the register so that we would not have to guess.

The register has one row per claim. Each row keeps the claim in the paper's words, our bar, our result and the rows behind it. It grew out of our own write-ups: each re-test was published with its data when it ran, and there are 29 write-ups so far.

Take one row. An adaptive chunking paper claims that choosing the chunk length from the entropy of sampled chunks follows the phases of the task, with long chunks in transport and short ones at the grasp. To choose one chunk length the method samples many chunks, and on our edge GPU this takes 4,751 ms. Our real-time budget is 298 ms. So the row records a fail on this rig, and it says nothing about the method on the hardware it was shown on.

Over the whole register, 12 of 37 claims hold and 18 fail. Several of the failures are not near misses. In those rows the learned method loses to a baseline with no learning in it, and we think this is the main lesson of the register.

2 How a row is made

Each row starts with a bar committed to version control before the test runs. We do this because a bar written after the result tends to drift toward it, even without intent. Section 5 checks from the commit history how well we kept this rule.

Before testing the learned method, we run the cheapest signal it has to beat, on the same rows. For a model that predicts the next state, this is copy-last: the next state is taken to be the current one. For a learned latent it is the commanded joints, because the state of the arm is already in them. For a judge of success it is the length of the episode, which on this rig already says a lot about the outcome. These baselines take minutes to run, and when a method loses to one of them, its bar no longer matters much.

We chose the papers for the stack this rig runs: world models, VLA policies, failure monitors, chunking and edge latency. So the failure rate describes this register, not the field. The register file lists every tested paper and every paper scheduled next, each entered before its test, so a paper cannot be dropped after its result is known.

Table 1: Verdicts by paper and by claim row (claim rows include our own baseline comparisons).

verdict	papers	claim rows
reproduces	4	17
fails	13	20
not proven	3	3
not tested	1	4
withdrawn	1	1
total	22	45

3 Results

Figure 1 summarises the register. Table 3 counts the verdicts by paper and by claim row, and the appendix gives every row.

Finding. 12 of the 37 claims taken from papers reproduce, and 18 fail. The other 8 rows test our own baselines, for example that the commanded joints beat the world model, and 5 of them hold.

At the level of papers, 4 of 22 reproduce and 13 fail.

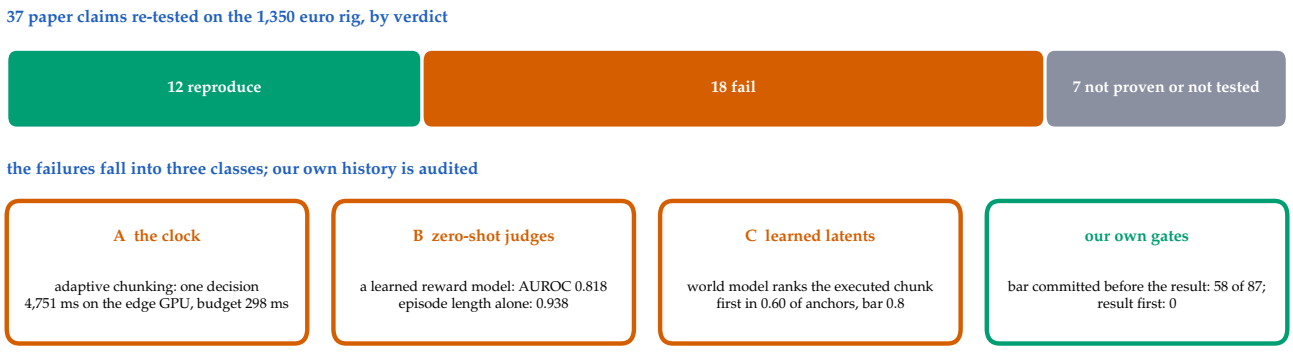


Figure 1: what survived when the papers left the lab: of the 37 paper claims, 12 reproduce and 18 fail; the failures fall into three classes (the clock, zero-shot judges against a trivial signal, learned latents against the commanded joints); our own gates have the bar committed before the result in 58 of 87 and the result first in 0.

4 Three failure classes

The adaptive chunking row fails on time. The method may work on a faster GPU, but on the edge GPU it misses the real-time budget. We call this class A, the clock. It is the easiest class to avoid, because a method can be timed on the target device before anything else is built.

Class B is about judges. Used without any training on this rig, a learned reward model detected failures with an AUROC of 0.818. The episode length alone reached 0.938. We put in class B every judge of success, failure or progress that loses to such a trivial signal. Our guess, which we have not tested, is that a zero-shot judge has never seen this rig, while the length of an episode is measured on it.

Class C is about learned latents. A world model ranked the executed chunk first in 0.60 of anchors, below its bar of 0.8, while an action-only baseline reached 0.68–0.90. Class C holds the latents used to plan, rank or track progress that lose to the joints, to copy-last or to an action-only baseline. On a small arm the commanded joints already describe much of the state, so a latent has to add something the joints do not have, and in these rows it did not.

These rows illustrate the classes and do not count them. The classes are our proposal, from the author's reading of the 20 failed claim rows, and a second reader is needed to know if they hold. For someone with a similar rig, they give an order for checking a paper: time it on the device, then compare it with the trivial baseline on the same rows, and only then look at its bar.

5 How far the register can be trusted

A verdict means something only if its bar came before its result. Notes and commit dates can be edited by hand. Commit ancestry cannot, unless one rewrites history that others have already cloned. So for each gate we find the first commit that contains its bar and the first that contains its result, and compare them (`repro/prereg-check/scripts/prereg_audit.py`, abridged):

Table 2: Commit-order class of each gate. The audited column is the one the paper uses; the first pass is what text search alone gave on the same gates.

class	first text pass	audited, 2026-10-31	audited, today
bar-first	36	58	61
same-commit	31	6	7
result-first	—	0	0
no bar	19	23	23
unresolved	1	—	—
gates	87	87	91

```
def ancestor(a, b):
    return subprocess.run(["git", "-C", str(RR), "merge-base", "--
      is-ancestor", a, b]).returncode == 0
...
            elif b[0] == res[0]:
                cls = "same-commit"
            elif ancestor(b[0], res[0]):
                cls = "bar-first"
            elif ancestor(res[0], b[0]):
                cls = "result-first"
```

Table 1 gives the class of each of the 87 gates that ran and were logged by 2026-10-31. Most are bar-first, and 0 are result-first. A same-commit gate is one whose bar and result entered the repository together, so it tells nothing about which came first. Text search alone first found 31 same-commit gates, mostly from batch logging. We went back to the earlier files that hold their bars, and 21 of them moved to bar-first and 4 to no-bar. The audit table in the repository gives the evidence for each move.

The early gates mixed point estimates and different intervals. We recomputed 10 of them from the stored rows with one statistic, the episode bootstrap (2,000 resamples, percentile 95 % CI). For 3 gates the bar falls inside the interval (Table 2), so we mark their verdicts as fragile but keep the pre-registered point verdicts. The false-

Table 3: The three gates whose verdict flips somewhere inside its 95 % interval.

gate	n	estimate	95 % CI	bar	point verdict
colour judge (no learning), accuracy	87	94.2 %	89.6–98.9 %	$\geq$ 95 %	fail
oscillation among failed episodes	25	24.0 %	8.0–40.0 %	$\geq$ 10 %	pass
judge threshold, false interventions	42	2.4 %	0.0–7.1 %	$\leq$ 5 %	pass

intervention threshold was fitted to meet its rate, so its interval is only descriptive.

All 29 write-ups rebuild from their published rows. The check for the cold reruns prints:

```
$ python3 repro/cold-reproduction/l0_check.py
ok  posts: 29 (report 29)
ok  L0 PASS: 29 (report 29)
ok  L1 REPRO: 6 (report 6)
ok  L1 DIFF: 0 (report 0)
ok  L1 classes: {'REPRO': 6, -'': 3, 'SKIP': 8, 'FAIL-ENV': 12}
L0 PASS
```

We ran the cold reruns from a fresh clone with no credentials. 6 gates reran and matched, 5 of them exactly. 12 more stop at the world-model environment, which is not published yet. The remaining 11 need a larger GPU, a private checkpoint or the arm, or a step the write-up leaves out. We also checked the citations and paper-side fields of the register against the papers' PDFs, and corrected them. The register names 38 arXiv ids. The ids missing from our mirror of the papers all belong to scheduled tests, and one tested paper has no arXiv id, so it could not be checked against a PDF.

6 Limitations

The largest risk is observer bias. One operator, the author, ran the rig and labelled every outcome, with the arm at rest but not always blind to it. Where the commit order is known, the bar predates the run, so a label could not move the bar. For 29 of the 87 gates the order is not known, and 21 bar-first verdicts rest on matching the bar text in an earlier file. A blind second reader would show whether verdicts flip. Under the bootstrap 3 verdicts are fragile, and the cold reruns cover 6 gates so far, because the world-model environment is not published yet.

7 Related work

Other work on this arm ranks policies. A real-world benchmark reports a taxonomy of failures and recoveries for VLA policies [13], and a fleet of SO-101 cells evaluates policies at scale [14]. In simulation, vla-eval runs VLAs across 14 benchmarks [8], and a benchmark of demonstration quality studies one method family [22]. We do not rank policies. We take the claim a paper makes about its method, fix a bar for it, and test it on one cheap rig. The closest work is RoboSeed [27]. It scores VLA and world-action models on five dimensions, among them reproduction evidence and deployment affordability. Its reproduction score counts audit milestones such as configuration mapping, task records and seeds, and its authors say the score is not a probability that a result is correct. A register answers a narrower question, whether a claim holds on one rig, and it gives the rows to check the answer.

8 Conclusion

Conclusion. We re-tested 37 claims from 22 robot-learning papers on a €1,350 rig with $45.10 of compute, and 12 held. Several methods lost to a baseline with no learning in it: copy-last, the commanded joints or the episode length. So our advice to anyone re-testing a paper on a cheap rig is to run that baseline first, on the same rows, with the bar committed before the run. Every row is published, and every number here can be rebuilt from the repository (`github.com/ravediamond/paper-register-reproducibility`). We will keep adding rows as the scheduled tests run.

References

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Appendix: the register

paper	their rig	claim tested	bar	our result	verdict
LeWorldModel [9]	sim	A small JEPA world model learns the arm's dynamics well enough to plan with it.	Dynamics: beat copy-last, every corruption ≥ 1.3 at $K = 30$. Planning: rank the executed chunk first among 8 in ≥ 0.8 of anchors.	dynamics 0.31 vs 0.89 copy-last (reproduces); planning top-1 0.60 vs 0.68–0.90 for an action-only baseline (fail)	fail
DINO-WM [1]	sim	A predictor on frozen DINOv2 patches plans better than one on a learned latent.	G1b top-1 ≥ 0.8 at some target distance.	arm-pose R^2 0.92 (vs 0.57 for W1); G1b top-1 0.55–0.61; no-model baseline 0.68–0.90	fail
RoboJEPA [20]	>$20k	Pretrained RoboJEPA predicts our latents, uses the actions, and runs on the robot's computer.	9-step L1 ≥ 15 % below copy-last (300M) and ≥ 5 % below shuffled actions; 2-view encoder ≤ 200 ms on the Orin.	11.8 % below copy-last (300M; 1B 15.8 % on 30 clips); 9–10 % below shuffled; encoder 526 ms on the Orin NX	fail
H-JEPA [19]	sim	A slow second level makes latent distance track task progress.	Progress monotonicity Spearman $\rho \leq -0.80$; false stalls ≤ 10 %.	level-2 ρ -0.73; false stalls 52 %	fail
D-JEPA [18]	<$5k	A decision-aligned ranking operator on a frozen world model picks better action chunks.	G1b regret ≤ 0.10 and $\leq$ the action-only baseline + 0.05.	regret 0.263 (W1 alone 0.263; action-only -0.007)	not tested
ContactGuard [16]	$5–20k	An event-anchored probe on imagined latents predicts grasp failure better than the current latent.	AUROC ≥ 0.85 and ≥ 0.05 above the current latent.	0.70 imagined vs 0.82 current	fail
FAIL-Detect and SAFE [2] [5]	>$20k	A conformal band on a learned failure score keeps false alarms at α while catching failures.	Successes flagged $\leq 1/20$; failures flagged ≥ 50 %.	22.7 % of successes flagged at $\alpha = 5$ %; 46 % of failures caught	fail
ReAP [21]	$5–20k	A risk head on an action-conditioned latent warns before failures.	$\geq 5/8$ failures caught at $\leq 1/20$ false interventions; action-conditioned iff true $\geq$ shuffled + 0.05.	v1 AUROC 0.93 came from time and length; v2 0.789 vs 0.708 shuffled, 3/8 at 5/42	fail
VLA-Corrector [17]	sim	A verifier on observable history flags a failing rollout early enough to correct it.	$\geq 5/8$ failures at $\leq 1/20$ false interventions.	the paper's verifier family on our rollouts (a GRU on pooled RGB statistics and actions): 2/8 caught at 3/42 false interventions, AUROC 0.59; cosine drift of the W1 latent 0/8, AUROC 0.30	fail
Robometer [7]	>$20k	A 4B reward model judges SO-101 rollouts zero-shot.	Pooled AUROC ≥ 0.85 and within-policy ≥ 0.8 on ≥ 5 of 7 policies.	pooled AUROC 0.818 on 2099 ArmnetBench rollouts; episode length alone 0.938	fail
SmolVLA [3]	<$5k	A 450M VLA succeeds on an SO-101 task and can run on low-cost hardware.	10-trial operator-judged gate (no numeric pass mark registered); chunk inference within the async budget (667 ms).	9/10 on bowls; 974 ms in PyTorch, 238 ms with vla.cpp on the Orin NX	reproduces

Table 4: The register, one row per paper, generated from the register file (internal model and gate names kept). The rig price of €1,350 counts the parts with a stated price; 2 small parts are not priced yet. Read the verdict column against the rig each paper was shown on.

paper	their rig	claim tested	bar	our result	verdict
TurboVLA [15]	sim	Our claim that the architecture made the policy ignore its instruction.	Task swap $\geq 5°$ and per-layout $\geq$ 75 % $\rightarrow$ our port was the bug; $<$ 1° with the control $\geq 0.095 \rightarrow$ the data.	upstream's own trainer on our data: task swap 0.020°, per-layout 38.9 %; control, their released LIBERO checkpoint run here: 0.105 ($\approx 5.5°$ of pan)	withdrawn
VLA-JEPA [6]	sim	VLA-JEPA trains cheaply on SO-101 data and runs on the robot.	10-trial operator-judged gate (no numeric pass mark registered); for reranking (H0): oracle / mean error $\leq$ 0.7 and N = 8 $\leq 1.5\times$ the latency.	9/10 for $9.7; 355 ms per chunk on the Orin NX; reranking no-go (0.76, 1.84×)	reproduces
REKI [23]	sim	Policies without memory oscillate between poses at a measurable rate.	$\geq$ 10 % of failed episodes oscillate (their criterion).	20.3 % of all episodes, 24.0 % of failures; REKI trained on my data: 19/20 + 19/20 on the arm, oscillates in 1/40	reproduces
Real-Time Chunking [4]	unstated	Conditioning a new chunk on the executing one removes the jump at chunk boundaries.	Best gradient-free method cuts the median jump $\geq$ 50 % against naive async, drift $\leq$ +10 %, within 667 ms.	prefix overwrite cuts the jump 71–73 % against naive async at +0.4 ms; tuned LeRobot identity mode is close	not proven
HWM (hierarchical latent world models) [10]	>$20k	A high-level world model over learned macro-actions beats the low-level model beyond about 1 s (their Fig. 6 crossover).	Crossover in [1.0, 2.0] s; at 2 s high-level error $\leq$ 0.95 × low-level; learned macro-actions $\geq$ net joint displacement on cosine.	no crossover: W1 0.382 $\pm$ 0.005 $\rightarrow$ 0.563 $\pm$ 0.007, high level 0.769 $\pm$ 0.009 $\rightarrow$ 0.759 $\pm$ 0.009 (0.73–2 s); cosine 0.608 $\pm$ 0.007 vs 0.551 $\pm$ 0.006	not proven
LeJEPA identifiability [12]	sim	A SIGReg JEPA trained on goal-directed data recovers the true latent state only partially and unevenly.	W1's mean R^2 from latent to joints $<$ 0.80; anisotropy (max $-$ min per-joint R^2) $\geq$ 0.20; the higher-R^2 model has the higher G1b top-1.	W1 mean R^2 0.565, anisotropy 0.639 (shoulder lift 0.91, wrist flex 0.27); frozen DINOv2-S 0.867 / 0.301	reproduces
SURVEYOR	sim	A threshold on the imagined-vs-reached latent distance accepts good plans and rejects bad ones ($\tau = 0.20$).	τ below the 10th percentile of non-equivalent pairs and $\leq$ 0.40; as a monitor $\geq$ 5/8 failures at $\leq$ 2/42; $\geq$ 0.05 AUROC over a shuffled drafter.	$\tau = 1.249$ on W1 latents of 69 real rollouts; as a monitor 3/8 at 6/42 (S = 30), AUROC 0.756 vs shuffled 0.560	fail
Adaptive Action Chunking (AAC) [11]	>$20k	Chunk length chosen from the entropy of N sampled chunks follows the task's phases (long in transport, short at the grasp), lowers the boundary jump, and costs little with batched sampling.	median h* in transport $\geq$ grasp window + 2; jump with AAC + overwrite $\leq$ fixed + overwrite at d = 8; an N = 20 decision $\leq$ 298 ms on the Orin.	h* 2 in transport and 2 at the grasp (86 % of decisions at the floor); jump 0.973 vs 0.940; N = 20 decision 4,751 ms (sequential) on the Orin	fail
PACT-VLA [24]	$5–20k	Picking among N sampled chunks by their flow residual under the pre-commit observation fits what happens next better than a random or stale pick, with no added control-step delay.	PACT beats a random pick and the stale-observation residual at d = 8 (CIs $<$ 0); its delay on the Orin $\leq$ the naive delay; at that delay it still beats a single sample.	offline $-0.27°$ vs random and vs the stale residual (both CIs $<$ 0); on the Orin 1,200 ms per cycle, 37 control ticks vs 8	fail
Horizon × depth (Euler steps vs execution horizon) [25]	>$20k	With long open-loop commitment, cutting a flow-matching VLA's Euler steps from 10 to 2 contracts the action cloud around an unchanged centroid, saves most of the inference time and avoids edge-GPU thermal throttling.	median R(2)/R(10) $\leq$ 0.70 with the centroid within 0.05 relative and 3°; N = 2 latency $\leq$ 0.85 × N = 10 on the Orin, N = 10 within 5 % of 238 ms.	R(2)/R(10) 0.78 (CI 0.73–0.81), centroid 0.018 and 2.35°; 196.1 vs 237.5 ms per chunk (0.83, vision 151 ms); no throttling at N = 10 on the Orin, at MAXN_SUPER or 25 W	not proven
REMORA [26]	>$20k	Reading a progress distribution out at the supported peak nearest the previous estimate steadies the estimate and lets it fall during real setbacks, with no worse detection of the stages.	Against per-frame argmax: premature not higher and missed $\leq$ +2 points (P1); flips halved, CI upper $\leq$ 0.7 (P2); a decrease inside $\geq$ 50 % of the reversal windows with $\leq$ 20 % false regressions (P3).	premature 0.22 vs 0.25, missed equal (P1 pass); flips 0.70 × argmax, CI 0.48–0.94 (P2 fail); setback recall 2/7, false 0.52 (P3 fail); MAE 0.184 vs median 0.218	fail

Table 4, continued.

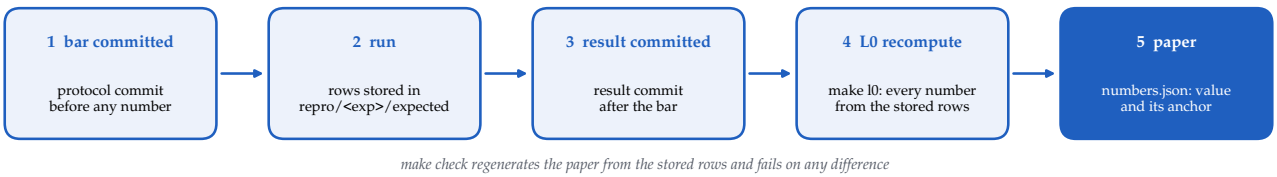


Figure 2: how every number in this paper is made: a protocol with its bar is committed before any number exists (1), the run stores its rows under repro/ (2), the result is committed after the bar and the commit order is audited (3), make l0 recomputes every number from the stored rows (4), make_paper.py writes numbers.json with each value's anchor and fills the text from it (5); make check regenerates the paper and fails on any difference.